

Kac_OS - LEFT manual

Version: 2026-09-18. A short DICE randomizes AS while preserving tempo, or selects the next TS engine. TS MIDI allocates up to three voices and responds to Note Off; physically held pads have priority.

KacOS is the K-Accumulator-inspired operating system for diceFriend. It combines carrier/modulator FM and phase modulation, self- and cross-feedback, wavefolding, an eight-node morph matrix, a delta-sigma step pattern, and Lorenz chaos. It is the standard-register version of KacOS: brighter and wider than KacOS Bass, with the same unstable sequenced architecture but without the two-octave bass transposition.

KacOS is designed for animated FM sequences, folded digital percussion, chaotic pitch walks, metallic strikes, morphing phase tones, touch-played three-voice gestures, and unstable melodic patterns that can lock to CV or MIDI clock.

KacOS has two main performance modes: **RS mode** and **Touch mode**. RS mode is the continuously running delta-sigma/Lorenz sequencer. Touch mode provides three playable voices and four timbre engines.

Main Operation

- **SHIFT** changes the function layer of the six knobs.
- **Short DICE in RS mode** radically randomizes the sound and delta-sigma pattern while keeping the current speed.
- **Short DICE in Touch mode** switches to the next Touch engine.
- **Hold DICE for about 3 seconds** switches between RS mode and Touch mode.
- **SHIFT + hold DICE for about 3 seconds** randomizes the current controls.
- **SHIFT + DICE, held for about 10 seconds** enters bootloader/update mode.

The knobs use pickup behavior after changing layers, modes, or randomized values.

CV and MIDI

- **CV1** works as a trigger/clock input and advances the delta-sigma/Lorenz system. In Touch mode it triggers the next available touch voice.
- **CV2** works as continuous modulation until pulses are detected; it then acts as a clock/trigger source. In Touch mode it can trigger allocated voices.
- MIDI Clock can take control of RS stepping when CV clock is not taking priority.
- Incoming MIDI notes directly set RS pitch or use three-voice allocation in Touch mode.
- In Touch mode, the per-voice pitch knob detunes an incoming MIDI note by approximately plus or minus one octave.
- Physical pads send MIDI notes on channels 1, 2, and 3.
- RS mode outputs the current generated pitch and, under external clock, an optional second transposed MIDI layer.
- RS activity is sent as channel aftertouch and continuous CV2 is sent as pitch bend.
- Musical incoming MIDI messages are passed through: notes, CC, program change, aftertouch, and pitch bend.
- Incoming realtime clock and transport are not echoed directly; clock output is regenerated by KacOS.

LEDs

- The SHIFT and non-SHIFT LEDs show the active knob layer.
- In RS mode, the four lower LEDs combine CV/audio activity with a binary indication of the current morph region.
- In Touch mode, touching pads lights the corresponding first three LEDs while the fourth shows output activity.
- After switching Touch engines, one lower LED briefly identifies the selected engine.
- DICE actions and mode changes produce confirmation flashes.

RS Mode: Delta-Sigma / Lorenz Sequencer

RS mode uses a mutable delta-sigma pattern as its pitch path. A Lorenz attractor biases the notes and adds controlled instability. That pitch drives a carrier/modulator pair whose sound moves across eight morph nodes: clean center, feedback phase modulation, two-operator PM, cross-PM, FM/AM, alternate feedback structures, and asymmetric PM.

The internal RS clock is slightly humanized. When KacOS is master, each step can move a little earlier or later depending on Depth, while MIDI Clock output remains organized as 24 PPQN pulses for downstream devices.

RS Controls

Normal layer:

- **Knob 1:** Internal speed, approximately 30-240 BPM; under external clock, transposes the second MIDI-output layer.
- **Knob 2:** Modulator harmonic ratio.
- **Knob 3:** FM/PM depth, Lorenz influence, and internal clock unevenness.
- **Knob 4:** Wavefolder amount.
- **Knob 5:** Harmonic stretch.
- **Knob 6:** Position in the eight-node morph matrix.

SHIFT layer:

- **Knob 1:** Fine modulator ratio/detune.
- **Knob 2:** Cross-feedback boost.
- **Knob 3:** Delta-sigma mutation chance.
- **Knob 4:** Delta-sigma pattern length, from 2 to 16 steps.
- **Knob 5:** Pattern smoothing, pitch-walk range, and Lorenz span.
- **Knob 6:** Output saturation.

RS Touch Pad Behavior

- **T1:** Freezes the delta-sigma pitch path while held.
- **T2:** Widens the pitch walk and Lorenz orbit influence.
- **T3:** Adds a controlled glitch/fold pressure without the old noise spike behavior.

Morph Matrix

Knob 6 moves continuously through eight synthesis regions:

- **Centre:** Clean carrier/modulator tone.
- **FBPM:** Self-feedback phase modulation.
- **2OP:** Two-operator phase modulation.
- **XPM:** Cross-coupled phase modulation.
- **FMNT:** Phase modulation combined with UFG amplitude movement.
- **FBPM2:** A more asymmetric feedback variation.
- **2OP2:** A denser two-operator region.
- **ASYM:** Strongly asymmetric phase modulation.

Intermediate knob positions interpolate between neighboring regions rather than switching abruptly.

Touch Mode

Touch mode provides three independent voices.

Normal layer:

- **Knob 1:** T1 pitch.
- **Knob 2:** T2 pitch.
- **Knob 3:** T3 pitch.
- **Knob 4:** T1 decay.
- **Knob 5:** T2 decay.
- **Knob 6:** T3 decay.

SHIFT layer:

- **Knob 1:** FM/PM depth.
- **Knob 2:** Phase grit/feedback.
- **Knob 3:** Modulator harmonic ratio.
- **Knob 4:** Wavefolder amount.
- **Knob 5:** Phase flip / ring-like inversion.
- **Knob 6:** Morph position for the KACC engine.

Incoming MIDI notes are allocated across the three voices. Pads have priority, so a voice currently held by a physical pad is not stolen by incoming MIDI. Note Off releases the matching voice. If MIDI notes stop arriving and a voice is left open, KacOS releases it after a short idle timeout to avoid stuck drones.

Touch Engines

Engine 0: KACC

The full RS-style morphing FM/PM core placed under a voice envelope. This is the broadest and most dynamic engine, moving from clean tones into feedback, folding, and asymmetric modulation.

Engine 1: Metal

An inharmonic modulator ratio with deeper phase modulation. It produces bell-like attacks, metallic overtones, and clangorous percussion.

Engine 2: Fold

A west-coast-inspired wavefolding voice with softer phase modulation. It works well for rounded notes that open into bright folded harmonics.

Engine 3: Grind

A heavier phase-feedback voice with a gritty, unstable edge. It is designed for growls, distorted hits, and chaotic sustained tones.

Performance Notes

- Standard KacOS sits higher and wider than KacOS Bass, with a useful carrier range from low tones into bright digital material.
- Knob 6 and Knob 3 form the main RS timbre pair: Morph selects the synthesis region and Depth determines how strongly it speaks.
- SHIFT Knob 5 controls pattern smoothness, pitch-walk range, and Lorenz span, so it can change phrasing and melody at the same time.
- Touch mode is reactive and does not auto-trigger: use pads, CV, or MIDI.
- External clocks replace the internal step timer and enable the second transposed MIDI-output layer.
- Short DICE in RS is intentionally broad: it randomizes the full musical/timbre state except tempo.